

1. Student Management System:

Class student

PRIVATE name, rollNo, Marks

Method acceptDetails()

Input name, rollno, marks

end method

method displayResult()

If calculateAverage() ≥ 40

print PASS

Else

print FAIL

end method

end class.

2. Library Book Management

Class Book

PRIVATE title, author, ISBN, isAvailable

Method issueBook()

isAvailable = FALSE

end method

method returnBook()

isAvailable = TRUE

End Method

method displayDetails()

print title, author, ISBN, isAvailable

End method

End class

Create b1, b2, b3

b1. issueBook()

b1. returnBook()

3. Employee Salary Calculation:-

```
class employee
    private empId, name, basicpay
    method calculategross()
        return basicpay + HRA + DA
    End Method
    method calculatenet()
        return calculategross() - PF - tax
    End method
    method displaypayslip()
        print calculatenet()
    end method
end class.
```

4. Shape Area Calculator:-

```
Class Shape
    Method CalculateArea()
end class
```

```
class Circle extends Shape
    Method CalculateArea()
        return  $3.14 * R * R$ 
    end class
```

```
class Rectangle extends Shape
    Method CalculateArea()
        return  $l * b$ 
    end class
```

```
class Triangle extends Shape
    Method CalculateArea()
        return  $0.5 * b * h$ 
    End class
```

Class Hierarchy:-

5. Vehicle Registration System:-

Class Vehicle

regNo, owner, fueltype

method displayDetails()

End class

class TwoWheeler extends Vehicle

engineCapacity

End class

class FourWheeler extends Vehicle

seatingCapacity, acStatus

End class.

6. Bank Account Hierarchy:-

Class Account Number

accNo, holder, balance

End class

Class SavingsAccount extends BankAccount

method withdraw(amount)

If Amount $\leq$ balance

balance -= Amount

End class

class CurrentAccount extends BankAccount

method withdraw(amount)

If Amount $\leq$ balance + overdraft

balance -= Amount

End class

ini Ver:

class

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End

class

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7. Hospital Patient Record System:-

Class Patient

private diagnosis, Prescription

Method generateBill()

End class

Class Inpatient extends Patient

Method generateBill()

Return days * dailycharge

End class

Class outpatient extends Patient

Method generateBill()

Return Consulting fee

End class.

8. E-Commerce Product Catalog:-

Class Product.

price

Method applyDiscount()

End class

Class Electronics extends Product

Method applyDiscount()

Return price * 0.90

End class

Class clothing extends Product

Return price * 0.80.

End class

Class grocery extends Product

Return price * 0.75

End class.

9. University Staff Hierarchy:-

```
class staff
```

```
    private salary
```

```
    protected name
```

```
    Method CalculateAllowance()
```

```
End class
```

```
class Teaching staff extends staff
```

```
End class
```

```
class Professor extends staff
```

```
End class
```

```
class Lecturer extends staff
```

```
End class
```

```
class NonTeaching staff extends staff
```

```
End class.
```

10. Smart Home Device Controller

```
class SmartDevice
```

```
    Method executeCommand(cmd)
```

```
end class
```

```
class SmartLight extends SmartDevice
```

```
    method executeCommand(cmd)
```

```
    Print "Temp Action"
```

```
End class
```

```
class SmartThermostat extends SmartDevice
```

```
    method executeCommand(cmd)
```

```
    Print "Light Action"
```

```
End class
```

```
class Smartlock extends SmartDevice
```

```
    private pin
```

```
    Method executeCommand(cmd)
```

```
    Print "Lock Action"
```

```
End class
```